# Thymosin Beta 4 sometimes called “TB-500 (Tβ4)”

### Acute Injury Speculative Dose: 2–3 mg SC twice weekly for several weeks, then 2–2.5 mg weekly

**Full-length thymosin beta 4 (Tβ4) is a naturally occurring 43-amino-acid peptide involved in actin regulation and tissue repair. Its actin-binding surface extends across much of the molecule, allowing it to sequester globular actin and participate in controlled actin exchange. It is studied for dermal wound repair, corneal surface recovery, cardiac protection, neurological repair, and hair-follicle support. Animal models show repair activity while human studies for local repair show mixed signals. Broader systemic injury recovery remains plausible. [1,3,5-17,26-29]**

💉 💉 **Probably safe:** Synthetic Tβ4 was well tolerated in a small intravenous study with daily dosing for 14 days. [3]

⭐ ⭐ **Probably effective:** Dermal and corneal repair are supported in animal injury models; human local-treatment benefit remains plausible with mixed trial outcomes. Tendon and broader systemic repair are plausible applications. [5-9,27-29,33]

**Cancer caution:** Tβ4 participates in angiogenesis and cell migration. Some tumor models show increased metastatic activity. A potential effect on an existing malignancy warrants caution, although findings differ by cancer type. [20,21]

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

Multiple small studies reported no or infrequent mild adverse events, including good hetpatic, renal, and cardiac safety. [3-4,18-19,25]

## Gray Market Confusion Between Tβ4 and TB-500

The names Tβ4 and TB-500 are used inconsistently in gray market and research-use-only sales and discussions. In published research, TB-500 is consistently identified as Ac-LKKTETQ, a synthetic N-acetylated seven-amino-acid peptide corresponding to residues 17–23 of full-length thymosin beta 4, while full-length Tβ4 contains 43 amino acids. These are distinct molecules, and findings for one should not automatically be attributed to the other. [31]

Vendor labels may use “TB-500” for either the fragment or full-length Tβ4. Labels such as “TB-500 (TB4)” or “TB-500 (frag)” can help indicate the intended identity, but the name alone does not establish what a sample contains. Ac-SDKP, the acetylated four-amino-acid peptide derived from Tβ4’s N-terminus, is another distinct molecule. [22,32]

Janoshik’s published reporting terminology distinguishes the full-length peptide from fragments as shown below. These are laboratory-specific labels, not universal product naming rules. [32]

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| Janoshik Label | Identity |
|---|---|
| **TB-500 (TB-4)** | Full-length, 43-amino-acid Tβ4 |
| **TB-500 889 (Da)** | The acetylated fragment corresponding to residues 17–23, Ac-LKKTETQ |
| **TB-500** | A fragment; consult the report's comments for its specific identity |

Always check the certificate of analysis (COA), reported molecular identity, and report comments rather than relying on a product name. [31,32]

## Full-length thymosin β4: animal injury-repair dosing and human-equivalent doses

Systemic rodent studies commonly used doses corresponding to human-equivalent-dose (HED) of approximately 30–70 mg per administration. Local treatments and some skin-wound experiments used substantially less.

| Study and injury model | Animal dosing protocol | HED for 70 kg | Findings |
|---|---|---|---|
| mouse fibular fracture [34] | 6 mg/kg IP on days 0, 3, 6, 9, 12; five doses | 34.1 mg | Greater fracture strength and stiffness |
| mouse hindlimb ischemia [35] | 150 µg/mouse, IM or IP, every 3 days for 3 weeks | ~34.1 mg | Increased angiogenic markers and reduced muscle fibrosis |
| mdx mouse muscle pathology [36] | 150 µg/mouse IP, twice weekly for 6 months | 28.4–42.6 mg | More regenerating fibers; no significant strength or fibrosis benefit |
| rat full-thickness skin wounds [9] | 60 µg/rat IP, days 0, 2, 4, 6 | ~2.3 mg | Improved re-epithelialization and wound contraction |
| rat skin wounds [37] | 60 µg/rat IP on injury day and every other day through day 7 | 1.9–2.7 mg | Histological evidence of improved wound healing |
| rat myocardial infarction [38] | 5.37 mg/kg IP on days 0–2; either stop or continue every third day through day 28 | 61.0 mg | Reduced infarct-related structural damage; limited improvement in ejection fraction |
| rat embolic stroke [15] | 6 mg/kg IP, starting 24 h after injury, then four more doses every 3 days | 68.1 mg | Improved neurological recovery and remyelination |
| rat traumatic brain injury [39] | 6 mg/kg IP, starting 24 h after injury, then four more doses every 3 days | 68.1 mg | Improved functional recovery and neural repair markers |
| rat traumatic brain injury [14] | 6 or 30 mg/kg IP at 6, 24, 48 h after injury | 68.1 / 340.5 mg | Both effective; larger dose generally more effective |
| rat stroke dose-response [40] | 2, 12, or 18 mg/kg IP, starting at 24 h and every third day for four additional doses | 22.7 / 136.2 / 204.3 mg | 2 and 12 mg/kg improved outcomes; 18 mg/kg did not show significant benefit |

| | | | |
|---|---|---|---|
| mouse myocardial infarction [41] | 1.6 mg/kg/day IP via continuous osmotic pump; one protocol started 7 days before injury and continued 7 days afterward; another started immediately after injury for 5 weeks | 9.1 mg/day | Reduced cardiac rupture and improved long-term remodeling/function |

### Most compelling systemic musculoskeletal protocol

The Brady fracture experiment used five 6 mg/kg injections spaced three days apart. Its HED was 34 mg per administration for a 70-kg adult. Bone calluses showed a 41% increase in peak force to failure and roughly 25% greater stiffness, measured 42 days after fracture [34].

### Evidence for much lower systemic doses

The original skin-wound work used approximately 2–3 mg HED [9]. The patent documentation reports reduced or absent activity at lower tested injection doses, including 30 µg and 6 µg IP compared with the 60-µg protocol. This establishes a lower effective exposure in a particular injury model, not a universal minimum dose [43].

### Nonlinear response

In the rat stroke dose-response study, 2 and 12 mg/kg were effective, but 18 mg/kg was not statistically effective. The authors' modeled optimum was an HED equivalent of approximately 42.6 mg. That optimum was calculated statistically rather than directly tested [40].

### Regeneration does not necessarily mean functional recovery

In the six-month mdx mouse study, Tβ4 increased regenerating muscle fibers without significantly improving muscle strength, cardiac function, or fibrosis. This limits the inference from cellular repair markers to clinical recovery [36].

The 2012 traumatic brain injury paper explicitly identifies its material as a synthetic copy of the naturally occurring 43-amino-acid peptide [14]. Many other studies specify Tβ4 and its commercial source, but do not independently document sequence verification.

### Animal dose conclusions

The animal literature supports three distinctions:

1. **Systemic repair experiments:** Many successful protocols cluster around 5–6 mg/kg in rodents, with **HEDs injections of approximately 34–68 mg** per 70-kg adult per administration [15,34,38,39].

2. **Local repair experiments:** Much smaller amounts can be biologically active at the injury site, including a single 1-µg surgical implantation with controlled release in the rat MCL model, which improved collagen organization and biomechanical strength at four weeks [42].


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3. **Dose and schedule are injury-dependent:** Results from skin, bone, cardiac, and nervous tissue do not establish one optimal systemic exposure for human tendon or ligament healing [9,34,38,40–42].

For evaluating Tβ4 specifically for tendon, ligament, or muscle tissue repair, I would give the greatest weight to 34 mg HED, every three days, for five applications. [34]

## Human Dosing Patterns

Ruff and colleagues studied intravenous doses of 42, 140, 420, and 1260 mg in four cohorts totaling 40 healthy volunteers. Participants first received a single dose of active treatment or placebo; after safety review, the assigned regimen was repeated daily for 14 days. These unusually large milligram doses are phase I safety testing, not a loading or maintenance protocol for injury recovery. [3]

No evidence-based human dose or dosing interval has been established for subcutaneous Tβ4. Community protocols commonly use a **higher-dose initial loading phase** followed by **less intense maintenance dosing**.

In absence of human testing for effective doses, an unresolved issue is why animal experiments suggest effective doses ten times larger than speculative community practices.

The following patterns are common community practices, speculative, and without evidence:

- **Mild injury and general recovery**: Loading, 2 mg twice weekly for 4 weeks. Maintenance, 2 mg once weekly during continued recovery.
- **Musculoskeletal injury**: Loading, 2.5 mg twice weekly for 4 to 6 weeks. Maintenance, 2 to 2.5 mg once weekly during continued recovery.
- **Severe injury or cardiac recovery**: Loading, 3 mg twice weekly for 6 weeks. Maintenance, 2.5 mg once weekly during continued recovery.

Sometimes twice-weekly dosing is divided into smaller daily doses.

## Common Stacking

Combining full-length Tβ4 with BPC-157 is an experimental repair strategy. Synergy is biologically plausible, but a 2026 rat tendon study of a commercial product labeled TB-500 found no additional benefit from the combination; the tested peptide sequence was not confirmed in the report, limiting attribution to full-length Tβ4. [33]

## Benefits

Tβ4 research developed from its role in the actin cytoskeleton into studies of wound repair, vascular growth, and cell survival. Experiments subsequently investigated cardiac injury, corneal damage, neurological injury, and hair growth. [3,5,6,9-17]

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The clinical record contains encouraging local-treatment signals, but also some unsuccessful primary endpoints. Small positive studies and secondary analyses support plausible repair benefits, but there are also negative findings. [5-8,26-28]

**Dermal wound repair:** Tβ4 accelerated reepithelialization and wound contraction in a rat wound model. [9] A human topical venous-ulcer trial reported a preliminary signal. [5] However the pooled endpoint did not establish overall efficacy. Local wound-healing benefit remains plausible. [29]

**Corneal surface repair and ocular comfort:** Topical Tβ4 is likely to support corneal repair based on animal injury studies. Human trials show selected improvements in ocular signs, comfort, or defect healing, but efficacy is mixed. The 72-patient dry-eye trial missed both primary endpoints while improving selected secondary endpoints; a separate nine-patient study reported improvements. [6,7,13]

In SEER-1, complete healing at four weeks occurred in 6/10 treated patients and 1/8 placebo patients, but missed statistical significance; a day-43 secondary healing endpoint favored treatment. [8] SEER-3 also missed its four-week primary healing endpoint. [27] Other studies are ongoing. [30]

**Tendon and ligament healing:** Benefits are plausible through cell migration, vascular support, and tissue remodeling. [2,9-11] A 2026 study of 32 rats found improved tendon failure load and selected histological outcomes with a commercial TB-500 product; only four tendons per group contributed to each endpoint class. Because the report did not confirm the tested peptide sequence, it cannot establish a molecule-specific effect for full-length Tβ4. [33]

**Skeletal muscle recovery:** Tβ4 plausibly supports recruitment of repair cells. Muscle-injury experiments showed endogenous Tβ4 expression and myoblast chemotaxis. [24] However, a separate mouse administration study did not enhance muscle regeneration. Faster or stronger muscle recovery therefore remains speculative. [23]

**Cardiac protection and repair:** Tβ4 is likely to support cell survival and repair after experimental myocardial injury, based on mouse studies and vascular mechanisms. A 2025 randomized recombinant-peptide trial in 96 STEMI patients found no significant overall infarct-size difference, with a favorable signal in the subgroup first treated within eight hours after PCI. Human cardiac benefit remains plausible. [10-12,26]

**Inflammatory signaling:** Tβ4 is likely to reduce selected inflammatory responses in injured tissues, supported by corneal, hepatic, and neurological animal studies. [13,14,18]

**Neurological tissue repair:** Tβ4 is likely to improve functional recovery and support neurorestorative processes based on rat traumatic-brain-injury and stroke studies. Mouse

demyelination studies also support oligodendrocyte maturation and remyelination. These are direct preclinical findings; human neurological recovery remains plausible. [14-16]

**Hair follicle support:** Tβ4 is likely to stimulate hair growth through follicular stem-cell migration and differentiation based on rodent experiments. Benefits for human scalp-hair loss are plausible. [17]

## Potential Applications

- **Acute soft tissue repair**. [5,9]
- **Chronic overuse injuries**. [2,9]
- **Ocular surface disorders**. [6-8,27,28]
- **Post infarction cardiac repair.** [10,26]
- **Neurological recovery**. [14-16]
- **Hepatic injury and fibrosis**. [18,19]
- **Renal tissue repair**. [25]

## Cautions

**Active or previous malignancy:** A plausible concern about supporting an existing tumor follows from Tβ4-related vascular growth and cell movement and from tumor-promoting experimental findings. Active cancer warrants particular caution and oncology involvement. Previous cancer requires assessment of current disease and recurrence risk. [20,21]

**Pregnancy and breastfeeding:** Avoid experimental use during pregnancy or breastfeeding as a precaution. [3,4]

**Children and adolescents:** The adult clinical regimens do not establish pediatric dosing or safety. Use in children or adolescents requires qualified clinical or research oversight. [3,4]

## Potential for Diminishing Benefit

Tβ4's intracellular actin-sequestering function differs from a conventional receptor-agonist mechanism. Clinical studies do not establish that cycling prevents diminished response. [2,9-11]

## Biological Mechanism

Full-length Tβ4 binds globular actin in a complex that prevents immediate filament incorporation. Exchange with profilin helps make actin available for controlled polymerization. These interactions support cytoskeletal changes involved in cell shape and movement. The molecule's extended binding interface is essential to understanding the full-length peptide. [2]

Tβ4 influences migration of repair-associated cells and vascular progenitors. Experiments link it to wound reepithelialization, collagen deposition, cardiac cell survival, and epicardial vascular


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responses. Improved blood supply during tissue repair is a plausible consequence of these vascular effects. [9-11]

Cardiac experiments identified a complex involving Tβ4, PINCH, and integrin-linked kinase, with Akt-associated survival signaling. [10] More recent recombinant-peptide experiments implicated ErbB2/Raf1 signaling and reduced pro-apoptotic Bad expression; ErbB2 inhibition abolished the protective cardiac response in mice. [26] Corneal and neurological models show tissue protection and modulation of injury-associated inflammation, while demyelination experiments implicate EGFR activation in oligodendrocyte maturation. [13-16]

The acetylated N-terminal residues of native Tβ4 contain the Ac-SDKP sequence. Laboratory and animal experiments demonstrate release through sequential processing involving meprin-α and prolyl oligopeptidase. Ac-SDKP formation provides a plausible contribution to some anti-inflammatory and antifibrotic effects. [22,25]

## Tβ4 and Cancer Risk

The cancer concern involves both repair biology and direct tumor-model findings. In a mouse melanoma experiment, increasing tumor-cell Tβ4 expression increased migration, tumor vascularization, and metastatic potential. This supports a context-specific tumor-promotion concern, but gene overexpression is not equivalent to administering an external peptide. [20]

The direction is not uniform. In multiple myeloma, increased Tβ4 expression reduced proliferation and migration in experimental cells and was associated with a protective pattern in mouse and patient-expression analyses. These contrasting results mean that neither universal cancer promotion nor universal protection is justified. [21]

The concern is most relevant to an existing or recurrent malignancy and should be assessed in its disease context. A claim that Tβ4 initiates cancer in humans is not established by these findings. [3,4,20,21]

## Conclusions

Human therapeutic benefit is application-specific and remains plausible rather than uniformly likely. [3-8,26-30] Tendon, ligament, general musculoskeletal, and systemic recovery applications remain plausible. Community loading and maintenance doses and patterns are unvalidated. [18-21,23,24]

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